

Mixing Operators on Weak-Topology Spaces Are Disjoint Mixing Along All Finite Families of Powers

Run fa_banach_001, lane 17 (agent_lane_17)

13 August 2026

Status: substantial partial answer, likely valid, pending human review. Question 3.4 is affirmative for every mixing operator on a topological vector space whose topology is weak. The general separable Fréchet-space question remains open.

1 Exact source question

Cardeccia and Muro ask in Question 3.4, arXiv PDF page 7 of [1]:

Source Question 3.4. *Is every mixing linear operator on a separable Fréchet space necessarily $\mathcal{AP}$ -hypercyclic? Equivalently, is every mixing operator multiply recurrent?*

Here multiple recurrence means that for every nonempty open set U and every $m \geq 1$, there is $n \geq 1$ such that

$$U \cap T^{-n}U \cap \cdots \cap T^{-mn}U \neq \emptyset.$$

- ii) B_ω is multiply recurrent;
- iii) $\prod_{l=1}^{n_k} \omega_l^{-1} e_{n_k} \rightarrow 0$ for some sequence $(n_k)_k \in \mathcal{AP}$ with the following property: given $p, m \in \mathbb{N}$ there exists $q \in \mathbb{N}$ such that the arithmetic progression of length m , common difference q and initial term $p + q$ is contained in $(n_k)_k$;
- iv) B_ω satisfies the $\mathcal{AP}$ -hypercyclicity criterion.

Recall that every hypercyclic backward shift is weakly mixing. Thus, every $\mathcal{AP}$ -hypercyclic backward shift is weakly mixing but the converse is not true. On the other hand, we will see in Theorem 4.1 that not every $\mathcal{AP}$ -hypercyclic operator is weakly mixing.

Recall also that a backward shift on a Fréchet space with basis is mixing if and only if $e_n \rightarrow 0$, thus every mixing backward shift is $\mathcal{AP}$ -hypercyclic but there are $\mathcal{AP}$ -hypercyclic backward shifts that are not mixing. In [30] there is an example of a mixing subshift in $\{0, 1\}^{\mathbb{N}}$ that is not $\mathcal{AP}$ -transitive and in [32] an example of a mixing linear operator such that $\{T, T^2\}$ is not d -transitive (but it is $\mathcal{AP}$ -hypercyclic because it is chaotic). Note also that if T is mixing then $T \oplus T^2 \oplus \cdots \oplus T^m$ is hypercyclic for every m . But we don't know the answer to the following.

Question 3.4. *Is every mixing linear operator on a separable Fréchet space necessarily $\mathcal{AP}$ -hypercyclic? Or equivalently, is any mixing operator multiply recurrent?*

The source proves the assertion for mixing backward shifts and for operators satisfying the strong Kitai criterion, but leaves the general implication open. We prove a different broad subcase.

2 Idea of the proof

On a weak-topology space, every basic open set imposes only finitely many linear-functional constraints. Shkarin's theorem says that mixing is exactly the absence of nonzero finite-dimensional invariant subspaces for the dual operator [2]. This turns the dual into a torsion-free module over $\mathbb{K}[t]$.

Given the finite-dimensional space L of functionals appearing in finitely many open sets, the generated module $\mathbb{K}[t]L$ is free. In free-module coordinates, elements of L have uniformly bounded polynomial degree. Multiplication by t^{jn} moves their supports into disjoint degree blocks when n is large. Therefore

$$L, (T')^n L, \dots, (T')^{mn} L$$

form a direct sum. The corresponding finite evaluation map on the original space is onto, so all conditions on $x, T^n x, \dots, T^{mn} x$ can be met exactly. This proves not only multiple recurrence but disjoint mixing of every finite family of powers.

3 The algebraic separation lemma

Lemma 1. *Let S be a linear operator on a vector space Y over $\mathbb{K}$. Suppose S has no nonzero finite-dimensional invariant subspace. Given a finite-dimensional subspace $L \subset Y$ and $m \geq 1$, there is N such that, for every $n \geq N$,*

$$L \oplus S^n L \oplus \dots \oplus S^{mn} L \tag{1}$$

is an algebraic direct sum.

Proof. No nonzero polynomial p can have a nonzero vector in $\ker p(S)$: otherwise the span of $y, Sy, \dots, S^{d-1}y$, where $d = \deg p$, contains a nonzero finite-dimensional invariant subspace. Thus the $\mathbb{K}[t]$ -module Y , with $t \cdot y = Sy$, is torsion-free.

The submodule $M := \mathbb{K}[t]L$ is finitely generated and torsion-free. Since $\mathbb{K}[t]$ is a principal ideal domain, M is free of finite rank. Choose a module basis. In these coordinates, every element of L is a vector of polynomials, and finite dimensionality gives a common degree bound d for all its coordinate polynomials.

Take $n > d$. If $y_j \in L$ and $\sum_{j=0}^m S^{jn} y_j = 0$, then in each free coordinate we have

$$\sum_{j=0}^m t^{jn} p_j(t) = 0, \quad \deg p_j \leq d.$$

The polynomial supports lie in the pairwise disjoint intervals $[jn, jn + d]$. Hence every p_j is zero, so every y_j is zero. This proves (1) with $N = d + 1$. $\square$

4 Weak-topology theorem

Recall that $T_1, \dots, T_m$ are *disjoint mixing* if, for every nonempty open $U_0, U_1, \dots, U_m$, there is N such that for every $n \geq N$ one can find $x \in U_0$ with $T_j^n x \in U_j$ for all j .

Theorem 1. *Let X be a Hausdorff topological vector space over $\mathbb{K}$ whose topology is weak, and let $T : X \rightarrow X$ be continuous linear and mixing. Then, for every $m \geq 1$, the operators $T, T^2, \dots, T^m$ are disjoint mixing. In particular, T is multiply recurrent.*

Proof. Fix nonempty open sets $U_0, \dots, U_m$. Because the topology is weak, for each j we may choose $u_j \in U_j$ and a basic weak neighborhood

$$V_j = \{x \in X : |f(x - u_j)| < \varepsilon_j \text{ for every } f \in F_j\} \subseteq U_j,$$

where $F_j \subset X'$ is finite. Let $L = \text{span}(\bigcup_{j=0}^m F_j)$.

By Shkarin's Theorem 1.1 [2], mixing of T implies that the dual operator T' has no nonzero finite-dimensional invariant subspace. Apply Lemma 1 to $S = T'$ and L . For all sufficiently large n , the linear map

$$A_n : X \longrightarrow (L^*)^{m+1}, \quad (A_n x)_j(f) = f(T^{jn} x) \quad (2)$$

is onto. Indeed, its algebraic transpose sends $(g_0, \dots, g_m) \in L^{m+1}$ to

$$g_0 + (T')^n g_1 + \dots + (T')^{mn} g_m,$$

which is injective by (1); since the target of A_n is finite dimensional, injectivity of the transpose is equivalent to surjectivity.

For each j , let $\alpha_j \in L^*$ be evaluation at u_j : $\alpha_j(f) = f(u_j)$. Surjectivity of (2) gives $x \in X$ such that

$$f(T^{jn} x) = f(u_j) \quad (f \in L, 0 \leq j \leq m).$$

Thus $T^{jn} x \in V_j \subseteq U_j$ for every j . This holds for every sufficiently large n , proving disjoint mixing.

Taking all $U_j = U$ gives multiple recurrence. □

Corollary 1. *Every hypercyclic continuous linear operator on $\omega = \mathbb{K}^{\mathbb{N}}$ with its product topology is $\mathcal{AP}$ -hypercyclic.*

Proof. The space ω is separable, complete, metrizable, and carries a weak topology. Shkarin's Corollary 1.2 [2] says that a hypercyclic operator on ω satisfies the same dual condition and hence is mixing. Theorem 1 makes it multiply recurrent. Proposition 2.1 of the source [1] identifies hypercyclicity plus multiple recurrence with $\mathcal{AP}$ -hypercyclicity. □

The supporting result used above is reproduced here for review.

We refer to [3, 6] for results on the structure of the set of hypercyclic vectors of operators on ω and to [4, 7] for results on hypercyclicity of operators on Banach spaces endowed with its weak topology. We characterize transitive and mixing operators on spaces with weak topology.

Theorem 1.1. *Let X be a topological vector space, whose topology is weak and $T \in \mathcal{L}(X)$. Then the following conditions are equivalent:*

- (1.1) 1) T' has no non-trivial finite dimensional invariant subspaces;
- (1.1) 2) T is transitive;
- (1.1) 3) T is mixing;
- (1.1) 4) for any non-empty open subsets U and V of X , there is $k \in \mathbb{N}$ such that $p(T)(U) \cap V \neq \emptyset$ for any polynomial p of degree $\geq k$.

Since ω is complete, separable, metrizable and carries weak topology, we obtain the following corollary.

Corollary 1.2. *Let $T \in \mathcal{L}(\omega)$. Then the following conditions are equivalent*

- (1.2) 1) T' has no non-trivial finite dimensional invariant subspaces;
- (1.2) 2) T is hypercyclic;
- (1.2) 3) T is hereditarily hypercyclic;
- (1.2) 4) for any sequence $\{p_k\}_{k \in \mathbb{N}}$ of polynomials with $\deg p_k \rightarrow \infty$, there is $x \in \omega$ such that $\{p_k(T)x : k \in \mathbb{N}\}$ is dense in ω .

5 Eight focused upgrade attempts

The following routes were checked after obtaining Theorem 1. They explain why the result is not promoted as a full answer.

1. **General dual separation.** Separating the diagonal orbit range from U^{m+1} produces a relation $\sum_{j=0}^m (T')^j f_j = 0$. For a norm or seminorm ball, however, the f_j range through an infinite-dimensional equicontinuous polar; finite module directness gives no uniform lower bound on these relations.
2. **Weak-topology module route.** This is the successful route above. It works because a basic neighborhood has a finite-dimensional polar span and yields the stronger disjoint-mixing conclusion.
3. **Full-sequence hypercyclicity criteria.** The strong Kitai criterion synchronizes all powers and is already handled by the source. Mixing does not imply that criterion; known mixing counterexamples to Kitai therefore prevent this from settling the general question.
4. **Quasi-rigidity.** Weak mixing supplies dense one-time recurrence along a common sequence, but $T^{n_k}x \rightarrow x$ does not imply $T^{j n_k}x \rightarrow x$: the operators T^{n_k} need not be uniformly bounded on the moving intermediate vectors.
5. **Composition-operator gluing.** Mixing runaway conditions provide small forward and backward images for one time. Pasting m orbit copies requires simultaneous control of overlaps and of pullback norms at $n, 2n, \dots, mn$; without bounded distortion these errors are not controlled.
6. **Direct-sum counterexamples.** Assigning a separate obstruction to each time n is compatible with mixing at the coordinate level, but an open ball in a separable direct sum cannot impose a uniformly positive obstruction on infinitely many tail coordinates.
7. **Scalar dual recurrences.** A fixed functional relation could forbid a three-term return near one vector, but any polynomially annihilated dual vector creates a finite-dimensional invariant dual subspace and is incompatible with mixing in the weak-topology model. No Banach-space replacement producing all-time obstructions was found.
8. **Known mixing non-Kitai constructions.** Grivaux-type and recent composition-operator and generalized-shift examples were inspected as counterexample candidates. Failure of Kitai's criterion does not imply failure of multiple recurrence, and no open set excluding all arithmetic return patterns could be extracted from their defining estimates.

The surviving obstruction is therefore precise: extend the finite-dimensional dual-block separation to the full equicontinuous polar of a Fréchet neighborhood, or build a mixing operator for which such a polar supports uniform near-relations that separate a diagonal orbit from U^{m+1} .

6 Verification, novelty, and scope

The proof is algebraic once Shkarin's characterization is accepted. The key checks are: torsion-freeness of the dual module; freeness of a finitely generated torsion-free module over the PID $\mathbb{K}[t]$; disjointness of shifted polynomial degree blocks; and the finite-dimensional transpose criterion for surjectivity. No compactness or Baire argument enters Theorem 1.

A bounded search checked the run’s four lightweight indexes, the parsed local arXiv corpus, the exact source question, Shkarin’s paper, and focused arXiv/web queries combining mixing, multiple recurrence, $\mathcal{AP}$ -hypercyclicity, weak topology, and ω . Recent linear-recurrence papers still treat multiple recurrence as a distinct notion; no explicit statement of the weak-topology/disjoint-powers theorem was found. The module argument may be folklore, so novelty confidence is moderate rather than definitive.

Human review should prioritize Lemma 1, especially the passage from torsion-free to free for $\mathbb{K}[t]L$, and the claim that injectivity of the transpose in (2) makes the finite-dimensional target map onto. The general normable or arbitrary Fréchet case is not claimed.

References

- [1] R. Cardeccia and S. Muro, *Multiple recurrence and hypercyclicity*, Math. Scand. **128** (2022), 517–532; arXiv:2104.15033.
- [2] S. Shkarin, *Mixing operators on spaces with weak topology*, Demonstratio Math. **46** (2013), 145–157; arXiv:1209.0979.